

Master Thesis Discussion (15th Meeting)

The learned smoother for geometric multigrid on surfaces

Method, error propagation, and the Stage I measurement

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Outline

1. **The question and the programme.** A classical solver that already works, and the narrower thing a learned smoother can therefore be asked for: the correction that *one node* of the existing cycle computes.
2. **The model problem.** The torus, the variable-coefficient Laplace–Beltrami problem, the manufactured solution, and the surface finite element discretisation.
3. **Multigrid, and how the error propagates.** The two-level correction, the identity $C_h = \Pi_F$, and the splitting $e = e_C + e_F$ that decides what each operation of the cycle is responsible for.
4. **The learned smoother.** The three available objectives, the one used and why, the norm, the component, the constraints — and what is deliberately left out of the loss and what that costs.
5. **Numerical experiments.** The classical reference, the Stage I measurement with its two capacity diagnostics, the answer, and one defect stated plainly.
6. **Conclusions.** What is established, the protocol a cycle result must be judged by, and the sequence of next experiments.

Starting Point, and the Question

- The starting point is a **classical, working solver**: the Laplace–Beltrami problem with a variable coefficient, discretised on the surface of a torus with high-order surface finite elements, solved by CG preconditioned by a geometric multigrid V-cycle.
- Both parts are implemented and verified. The discretisation converges at the optimal rates for the degree used, and the preconditioned iteration is **mesh-independent** — it reduces the residual by 10^{-10} in essentially the same number of steps whether the system has one hundred or ten thousand unknowns.

A learned smoother is therefore *not* asked to repair a solver that fails. It cannot distinguish itself by restoring scalability, because scalability is already there. What it can be asked to do is narrower: reduce the *constant*.

The question this work answers

Not whether a network can solve the surface problem, and not whether it can replace the multigrid solver. The question is whether a network can supply the correction that one node of the existing cycle computes.

- The **smoothing node** is the natural place to ask this, for a structural reason rather than an opportunistic one: of smoothing, residual evaluation, restriction, the coarse solve and prolongation, the relaxation sweep is the *only* operation whose replacement leaves the multilevel structure intact.
- Everything else keeps its classical form, so a change in the cycle's behaviour is **attributable to the substituted node** rather than to a re-engineering of the method — and that literature has been shown to suffer from weak baselines and reporting bias [9].

The work is organised as three stages, named from the beginning so that a Stage I result is not read as more than it is: **I** — one level, does the learned step reduce the components the coarse grid cannot reach? (*measured; this talk.*) **II** — what does it do to the components the coarse grid *does* reach? (*designed.*) **III** — does the substitution survive the recursive V cycle? (*protocol fixed in advance.*)

The Surface and the Equation

Let $\Gamma \subset \mathbb{R}^3$ be the surface of a torus with major radius $R = 2$ and minor radius $r = 1$, parametrised by ξ (around the ring) and η (around the tube),

$$x(\xi, \eta) = \begin{pmatrix} (R + r \cos \eta) \cos \xi \\ r \sin \eta \\ (R + r \cos \eta) \sin \xi \end{pmatrix}.$$

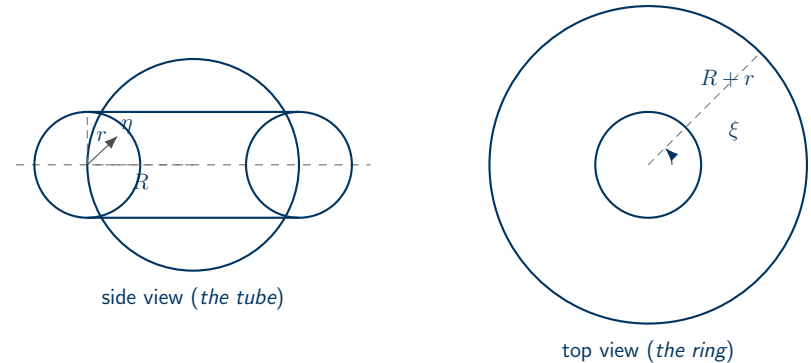
The induced metric is diagonal in these coordinates, with scale factors

$$h_\xi(\eta) = R + r \cos \eta, \quad h_\eta = r, \quad dS = h_\xi h_\eta d\xi d\eta.$$

We solve the Laplace–Beltrami problem with a variable coefficient and a reaction term,

$$-\nabla_g \cdot (\kappa \nabla_g u) + u = f \quad \text{on } \Gamma.$$

- Γ is *closed*: no boundary, so no boundary condition and no boundary data.
- The reaction term makes the operator coercive — on a closed surface the pure Laplace–Beltrami operator has a one-dimensional null space of constants — and it makes the discrete system positive definite for every coefficient considered.



The scale factor $h_\xi = R + r \cos \eta$ makes the metric **non-constant**: cells near the inner equator are shorter in the ξ direction than cells near the outer equator. This is the geometric reason the problem is not translation-invariant and the classical symbol analysis of smoothing does not apply.

Manufactured solution, and the discretisation

$\kappa(\xi, \eta) = 1.1 + \sin^2 \xi \cos^2 \eta$, $u(\xi, \eta) = \sin \xi \sin \eta$, so $\kappa \in [1.1, 2.1]$; u is known in closed form, so every run yields a genuine discretisation error, and $f = -\nabla_g \cdot (\kappa \nabla_g u) + u$ is evaluated **analytically** — the residual is exact up to round-off. Surface elements [3, 4] in deal.II [5]: FE_Q<2,3>(p) with MappingQ<2,3>(p) and QGauss<2>(2p+1) on a TorusManifold<2>, uniformly refined. One shared cell worker assembles the active system and the level matrices, so the hierarchy is a true geometric hierarchy for this operator.

The Hierarchy, and One Two-Level Step

Uniform refinement of the torus mesh gives a nested sequence of finite element spaces $V_0 \subset V_1 \subset \dots \subset V_L$, with $\dim V_\ell = n_\ell$ and $h_\ell = h_0 2^{-\ell}$. Between consecutive levels we use the canonical prolongation and the transposed restriction, with the Galerkin coarse operator:

$$P_\ell = P_{\ell \leftarrow \ell-1} : \mathbb{R}^{n_{\ell-1}} \rightarrow \mathbb{R}^{n_\ell}, \quad R_\ell := P_\ell^\top, \quad A_{c,\ell} = P_\ell^\top A_\ell P_\ell.$$

Since A_ℓ is SPD and P_ℓ has full column rank, $A_{c,\ell}$ is SPD as well, and the coarse problem is the restriction of the fine problem to the coarse space rather than an independently assembled operator. For the rest of this part the level index is fixed and dropped: $A := A_\ell$, $P := P_\ell$, $A_c := P^\top A P$.

One two-level step is: *relax, form the residual, restrict it, solve the coarse problem for the correction, prolongate and add, relax again*. Writing $\tilde{u}$ for the current approximation, $r = b - A\tilde{u}$ for the residual and $e = A^{-1}r = u - \tilde{u}$ for the algebraic error, the coarse correction e_H solves $A_c e_H = Rr$, so

$$\tilde{u}^{\text{new}} = \tilde{u} + P A_c^{-1} R r, \quad e^{\text{new}} = e - P A_c^{-1} R A e = \underbrace{(I - P A_c^{-1} R A)}_{=: C_h} e.$$

C_h is what one two-level step does to the error, in one line. The next frame says it is not merely *close to* a projection: with the exact coarse solve and the transposed restriction used here, it *is* exactly the operator the error splitting is built from — and that is why the analysis that follows is possible at all.

The Correction Is the Complement Projector

Proposition

Let $A_c = P^T A P$ be the Galerkin coarse operator and let $R = P^T$. Then

$$P A_c^{-1} R A = P (P^T A P)^{-1} P^T A = \Pi_C, \quad \text{and hence} \quad C_h = I - \Pi_C = \Pi_F,$$

where Π_C is the A -orthogonal projection onto $\text{range}(P)$ and $\Pi_F = I - \Pi_C$ its A -orthogonal complement.

Proof. The middle factor is precisely the stated expression. $\Pi_C^2 = \Pi_C$ follows by cancelling $(P^T A P)^{-1} (P^T A P)$, and $\Pi_C^T A = A \Pi_C$ is immediate from the symmetry of A and A_c . ■

Two things are worth stating plainly.

- It is an **identity, not an approximation**: it holds exactly and is checkable in the same code that produced the experiment, with no new data.
- It is the reason the two-level analysis below is possible at all — because $C_h = \Pi_F$ exactly, the two-grid error propagator is

$$E_{\text{TG}} = S_{\text{post}} (I - P A_c^{-1} R A) S_{\text{pre}} = S_{\text{post}} \Pi_F S_{\text{pre}},$$

with no error term to account for. Anything in the cycle that is not a smoothing sweep is a projection, and a projection is something one can reason about exactly.

The identity uses an *exact* solve of the coarse problem. In practice the coarsest level is small enough to be solved directly, so this is the configuration the solver actually runs. Where the recursion continues to further levels, the two-grid factor becomes the appropriate bound rather than an identity — the classical distinction between a two-grid analysis and a V-cycle analysis [1], and exactly the gap Stage III has to close.

The Error Splits, and What the Split Assigns

Every algebraic error splits uniquely and A -orthogonally,

$$e = e_C + e_F, \quad e_C = \Pi_C e \in \text{range}(P), \quad e_F = \Pi_F e,$$

characterised by three identities, all of which hold exactly:

$$e_C^\top A e_F = 0, \quad P^\top A e_F = 0, \quad \|e\|_A^2 = \|e_C\|_A^2 + \|e_F\|_A^2$$

The *second* identity carries the interpretation. Since $\Pi_F = I - P(P^\top A P)^{-1} P^\top A$, applying $P^\top A$ to e_F gives zero: e_F is exactly the part of the error the coarse grid **cannot see**. The correction, which acts by $P A_C^{-1} P^\top$ on the residual, is identically blind to it — equivalently, the residual produced by e_F is orthogonal to the coarse space.

$\mathbb{R}^n = \text{range}(AP) \oplus \ker(P^\top)$, of dimensions n_c and $n - n_c$. A complement residual satisfies $P^\top r_F = 0$; a coarse residual lies in $\text{range}(AP)$; and the two subspaces meet only at the origin. So a step trained to act on e_F *has never been shown a single input from $\text{range}(AP)$* , and its behaviour there is a pure extrapolation — whether that is harmless or harmful is a measurement, not an assumption.

So the smoother's duty is not “reduce the error”. It is narrower: e_C is the coarse-grid correction's business, and the correction is identically blind to e_F , so **if the smoother does not reduce e_F then nothing in the cycle does**. That is what makes the first experiment an *extrapolation test* rather than a memorisation test, and it is the vocabulary in which the defect found there is stated. The classical smoothers are the right comparison set: they too are judged by what they do to the whole space, and the split says which part of that matters.

Where a Learned Step Enters, and What It Costs

Relaxation propagates the error by $e^{(k+1)} = (I - BA)e^{(k)}$, reducing the oscillatory part quickly and the smooth part slowly; the coarse-grid correction does the opposite. The learned step keeps the additive form and replaces the fixed matrix B by a network,

$$v^+ = v + \delta v, \quad \delta v = B_\theta(Ae, X), \quad S_\theta = I - B_\theta A,$$

where X carries the geometric information (the coordinates of the degrees of freedom on the surface) and θ are the weights. Substituting this for one or both sweeps in $E_{\text{TG}} = S_{\text{post}}\Pi_F S_{\text{pre}}$ makes the two-grid stage *predictable* rather than exploratory:

One-sided use $S_{\text{post}} = S_\theta$

$E_{\text{TG}} = S_\theta \Pi_F$, so $\|E_{\text{TG}}e\|_A = \|S_\theta e_F\|_A$. Measured in the complement energy, the two-grid factor is then **exactly** the Stage I quantity ρ_F — the single-level measurement is a *direct prediction* for this configuration, not an analogy.

Symmetric use $S_{\text{pre}} = S_{\text{post}} = S_\theta$

$E_{\text{TG}} = S_\theta \Pi_F S_\theta$. Here S_θ first acts on the *full* error, and $S_\theta e_C$ generally acquires a component outside $\text{range}(P)$ which Π_F then keeps. Define the **leak** $\lambda := \mathbb{E}[\|\Pi_F S_\theta e_C\|_A / \|e_C\|_A]$.

A smoother that mapped $\text{range}(P)$ into itself would have $\lambda = 0$ and would cost the cycle nothing; a smoother that does not, does not. The leak is the quantity Stage I did *not* measure — though its data determines it — and it is the first thing Stage II has to compute. It is also the right way to read the defect reported later: a measured growth of the coarse component is an **upper bound** on λ , not evidence that $\lambda > 0$, because a step that amplified e_C purely *within* $\text{range}(P)$ would have $\mu_C > 1$ and $\lambda = 0$, and would cost the cycle nothing at all.

Three Objectives, and Which One Is Used

A classical smoother is fixed by a few discrete choices — a scheme, a damping factor, a sweep count — and analysed through the propagator $S = I - BA$. A learned smoother is fixed instead by the weights of a network, and a network is not a scheme. The decision that replaces the choice of scheme is therefore the choice of *objective*, and it is not unique.

Objective	What is minimised	What it buys, and what it costs
1. Correction matching	$\frac{1}{N} \sum_j \ \mathcal{C}(\kappa, r^{(j)}) - Br^{(j)}\ ^2$	Regression onto an operation that is already implemented, so the easiest to train — and it <i>cannot exceed the smoother it imitates</i> .
2. Error damping	$\frac{1}{N} \sum_j \ e^{(j)} - \mathcal{C}(\kappa, Ae^{(j)})\ ^2 / \ e^{(j)}\ ^2$	Trains the network to <i>reduce the error</i> rather than to imitate an operator. This is the objective used here.
3. Cycle unrolling	$\ \mathcal{V}_\theta(0) - A^{-1}b\ $	Optimises the quantity ultimately of interest through a differentiable V-cycle. Most aligned with the goal, <i>hardest to interpret</i> : a gain cannot be assigned to any one node.

Objective 3 is what a complete solver would be trained against, and Stage III will have to answer for it [6]. Objective 2 is what Stage I can *interpret*: it measures *one step*, so a result is attributable to that step and not to a composition.

The choice is a modelling decision, not a technicality. Objective 2 is also the only one of the three that can be scored on the same test errors, in the same norm, for a network and for Jacobi, Gauss–Seidel and SSOR alike — which is what makes the classical baseline fair.

The Objective Used, and What Is Left Out of It

The choice of error class $\mathcal{H}$ is the substance of the objective. $\mathcal{H}$ must contain the errors that actually arise in the cycle — oscillatory components on each level, errors produced by restriction, and, on a surface, errors whose local size varies with the coefficient and the metric — and it must contain errors *not* used in training, since a small value on the training sample bounds the average behaviour on that sample and nothing more. The norm is the discrete *energy* norm, because that is the norm in which the iteration contracts.

Samples are generated **algebraically**, so no exact solution of the PDE is ever required: an error $e^{(j)}$ is drawn, an artificial residual $r^{(j)} = Ae^{(j)}$ is formed, the network returns $\delta v^{(j)}$, and the remaining error is $\tilde{e}^{(j)} = e^{(j)} - \delta v^{(j)}$.

Manufacturing the residual from a *known* error means the target is available in closed form and every method in the comparison can be scored on the same test errors in the same norm. Five generators — smooth, multiscale, localized, algebraic, mixed — each on its own random stream, so train, validation and test cannot share a draw.

The price of leaving the coarse term out is measured, not hidden: $\mu_C = 1.0400$ at width 256, 1.0928 at width 768; *every* coarse sample is grown, by up to 19.7%.

Which component the loss trains on

$$\mathcal{L}_{\text{smooth}}(\theta) = \frac{1}{N} \sum_{i=1}^N \frac{\|e_{F,i} - B_{\theta}(Ae_{F,i}, X)\|_A^2}{\|e_{F,i}\|_A^2}$$

Only the complement is used — as the target *and* as the input. Nothing in this loss mentions e_C . That is not an omission: training on e_F alone prevents the optimiser from reducing the loss by acting on the coarse component, which is the coarse-grid correction's job and is not present in this stage at all.

Two constraints put the network in the same class as the objects it is compared against. **Homogeneity of degree one** in r : every classical smoother is linear and therefore exactly homogeneous, the exact solve $\delta e = A^{-1}r$ being the extreme case. **The zero-residual fixed point**: the branch is bias-free, so $\mathcal{C}(\kappa, 0; \theta) = 0$ holds structurally — otherwise an exact discrete solution would not be a fixed point and the cycle would acquire an error floor no number of iterations removes.

The Classical Reference: the Baseline Is Not Weak

Establishing that is the first result rather than a preliminary. Cubic (Q_3) discretisation, four uniform refinement cycles, outer iteration to a relative residual tolerance of 10^{-10} :

Cycle	Cells	Levels	DoFs	CG iterations	H^1 error	L^2 error	L^∞ error
0	16	1	144	1	5.856e-02	1.617e-02	5.417e-03
1	64	2	576	12	8.054e-03	1.114e-03	4.698e-04
2	256	3	2304	13	1.035e-03	7.225e-05	3.394e-05
3	1024	4	9216	14	1.302e-04	4.567e-06	2.175e-06

The discretisation converges at the expected rates.

Halving the mesh size divides the H^1 error by 7.27, 7.79, 7.94 and the L^2 error by 14.5, 15.4, 15.8 — observed orders $2.86 \rightarrow 2.99$ and $3.86 \rightarrow 3.98$, the optimal $\mathcal{O}(h^p)$ and $\mathcal{O}(h^{p+1})$ for cubic elements.

The multigrid preconditioner is doing its job. The iteration count barely moves — 12, 13, 14 — while the mesh is refined three times and the system grows sixty-fourfold, from 144 to 9216 unknowns. This is mesh-independence. (Cycle 0 has a single level, so its coarse solve is exact and CG terminates in one step; it is not evidence either way.)

The constraint this puts on any claim

A learned smoother cannot distinguish itself by fixing a failure of scalability that does not occur. What it can be asked to do is reduce the *constant*: fewer iterations to the same tolerance, or the same count at lower cost per iteration. The reverse also holds, and it is part of the result — a learned smoother that shows no gain here has not been shown useless, because the baseline it failed to beat was not weak.

The Stage I Experiment, and What Is Asserted Rather Than Assumed

Stage I asks one question and answers it with one table: *does the learned step, applied to one level, reduce the error components the coarse grid cannot reach — preferentially, and better than the classical relaxations?* It is scored by the per-sample reduction factor and its mean, $\rho_F(e_F) = \|e_F - B_\theta(Ae_F, X)\|_A / \|e_F\|_A$ and $\mu_F = \mathbb{E}[\rho_F]$, with μ_C defined identically on the same test errors; selectivity is $\mu_F \ll \mu_C$. Median, sd, p95, maximum and $\mathbb{P}(\rho > 1)$ are recorded beside every mean, because a mean cannot distinguish a method that reduces everything a little from one that reduces most samples a lot and amplifies a few.

Item	Setting
Hierarchy	six levels, L_0 with 16 unknowns through L_5 with 16384, uniformly refined. Level used: $\ell = 3$, $n = 1024$ unknowns, coarse $n_c = 256$, <i>complement dimension</i> 768
Coarse operator	Galerkin $A_c = P^T A P$; P is 1024×256 , 9 nonzeros per column, row sums exactly 1
Data	five generators, splits 2048/512/1024
Model	branch MLP, depth 4, bias-free, <i>width the only variable</i> ; trunk frozen orthonormal Fourier, $K = 16$
Schedule	batch 256, lr 3×10^{-3} , 1200 epochs, patience 300; seed 20260925 applied <i>before</i> the model is constructed

Every statement the experiment rests on is **asserted by the program** on its smoke test and re-measured on the actual data splits. Nothing is assumed:

Identity	Rel. residual	Where
$e_C^T A e_F = 0$	1.4×10^{-15}	projector
$P^T A e_F = 0$	2.6×10^{-14}	projector
$\ e\ _A^2 = \ e_C\ _A^2 + \ e_F\ _A^2$	4.4×10^{-15}	projector
$\lambda_{\min}(A_c) > 0$	0.274290	coarse op
$\max \ P^T r_F\ / \ r_F\ $	2.5×10^{-14}	splits
$\min \ P^T r_C\ / \ r_C\ $	1.02	splits

The last two rows confirm the direct sum on *real data*: training inputs lie in $\ker(P^T)$ to round-off, coarse residuals do not, and the two input sets therefore meet only at the origin.

Two Capacity Diagnostics, Reported Beside Every Number

A DeepONet output has the bilinear form $\delta e = \sum_k b_k(r)T_k(x)$, so the correction lies in $\text{span}(T)$ *whatever the branch returns*. A small μ_F can therefore be a statement about the trunk rather than about smoothing — and the first run of this experiment measured exactly that. **The floor is computable before training.**

1. The trunk sets a floor

$$\text{reach}_F(e) = \|\text{proj}_{\text{span}(T)} e_F\|_A^2 / \|e_F\|_A^2,$$

$$\mu_F \geq \mathbb{E}[\sqrt{1 - \text{reach}_F}].$$

Trunk	dim	rank	floor on μ_F
Fourier $K = 2$	48	24	0.997784
Fourier $K = 6$	336	168	0.965419
Fourier $K = 10$	880	440	0.786705
Fourier $K = 12$	1248	624	0.563591
Fourier $K = 14$	1680	840	0.286669
Fourier $K = 16$	2176	1023	0.000675

The reason is structural, not numerical. $\text{range}(P)$ is what the coarse lattice can represent (16×16 , Nyquist 8), so its A -orthogonal complement is essentially the band *between the coarse and the fine Nyquist* — modes $8 \lesssim |k| \leq 16$. A Fourier trunk needs modes up to 16. The headline run therefore *freezes* the trunk at $K = 16$: then $\text{span}(T) = \mathbb{R}^n$, $T^\top T = I$, and the floor is **zero** — provably non-binding.

2. The branch sets a second floor

The branch narrows to *width* units, so its Jacobian factors through $\mathbb{R}^{\text{width}}$, while the target of this stage is a *linear map of rank 768*. A branch narrower than that cannot express the exact correction, whatever the optimiser does. The reference curve is the best *linear* map of rank $\leq k$ fitted to the same pairs with the same weighting — a property of the *data*, no network is trained.

rank k	μ_F	μ_C	μ_C / μ_F
64	0.8057	1.0103	1.25
128	0.7160	1.0150	1.42
256	0.5860	1.0200	1.74
384	0.4768	1.0224	2.14
512	0.3666	1.0237	2.79
640	0.2375	1.0245	4.31
768	0.0000	1.0249	∞

The curve saturates at the complement dimension — the rank a branch *must* have to do the job. And $\mu_C \approx 1.02$ at *every* rank: selectivity is not something the network has to discover. The interesting question is therefore not whether selectivity appears but how well the complement is reduced — and a branch of width w sits near the $k = w$ row, so a result *at* that row measures the width, not the optimiser.

The Measurement, and the Answer

1024 test errors, level $\ell = 3$, $n = 1024$, complement dimension 768. Every method is scored on the *same* test errors in the *same* A -energy norm. The width-768 run reached its best validation $\mu_F = 0.1701$ at epoch 1199 of 1200, in 2092 s.

Smoother	μ_F	μ_C	$\max \rho_F$	$\mathbb{P}(\rho_F > 1)$	median	sd	p95	μ_C/μ_F
Jacobi	0.6996	0.6734	1.4310	0.1387	0.6653	0.2598	1.3035	0.96
Damped Jacobi	0.4248	0.7380	0.7230	0.0000	0.4064	0.0592	0.5522	1.74
Gauss–Seidel	0.3664	0.6180	0.5658	0.0000	0.3511	0.0476	0.4513	1.69
Symmetric GS	0.1889	0.4695	0.3848	0.0000	0.1743	0.0397	0.2606	2.49
SSOR	0.1889	0.4695	0.3848	0.0000	0.1743	0.0397	0.2606	2.49
DeepONet (width 768)	0.1456	1.0928	0.3198	0.0000	0.1334	0.0886	0.2695	7.50
A^{-1} (reference)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	—
ridge (diagnostic)	0.0000	1.0249	0.0000	0.0000	0.0000	0.0000	0.0000	∞

Answer: yes — and it is the best single-step method in the table

With a branch wide enough to represent the complement correction, $\mu_F = 0.1456$, $\mu_C = 1.0928$, $\mu_C/\mu_F = 7.50$, which places it first of the six methods on complement reduction: DeepONet $0.146 < \text{SymGS } 0.189 < \text{GS } 0.366 < \text{damped Jacobi } 0.425 < \text{Jacobi } 0.700$. It removes 85% of the complement's A -energy in one step — better than symmetric Gauss–Seidel — and *no sample is amplified* ($\mathbb{P}(\rho_F > 1) = 0$, worst case $\max \rho_F = 0.320$).

SSOR and symmetric Gauss–Seidel coincide because $\omega = 1$ is symmetric GS for this sweep; the table shows them agreeing to every printed digit, which is a check on the implementation rather than a coincidence. The network and damped Jacobi both have a free parameter selected on the *same* validation metric and scored on test, so the comparison is symmetric. The ridge and rank- k rows are *diagnostics, not competitors*: without them there is nothing to say whether $\mu_F = 0.15$ is good, bad, or exactly what the architecture permits.

Per Mechanism, and a Defect Worth Naming

μ_F by the generator that produced the error.

Mechanism	w256	w768	Damp. Jac.	GS
smooth	0.0907	0.0129	0.4554	0.4029
multiscale	0.6144	0.0978	0.4345	0.3711
localized	0.4270	0.0627	0.4461	0.4155
algebraic	0.7675	0.2503	0.3961	0.3325
mixed	0.6856	0.1853	0.4166	0.3523

At width 256 the network looked strong on smooth errors and weak on algebraic ones, which suggested the learned and classical steps fail in *opposite regimes* and ought to be combined. **That reading does not survive the wider run.** At width 768 the network beats Gauss–Seidel on *every* mechanism — including the purely algebraic coefficient-space noise it appeared worst at, by 0.2503 against 0.3325. There is no complementarity to exploit; there was an under-capacity branch and a mechanism whose errors it could not represent.

What this predicts for Stage II. $\mu_C = 1.0928$ is an *upper bound* on the leak λ , not evidence that $\lambda > 0$: a step that amplified e_C purely within $\text{range}(P)$ would have $\mu_C > 1$, $\lambda = 0$, and would cost the cycle nothing. What *is* established is comparative and harder to dismiss — the learned step is the only one that grows e_C at all, and it grows it *more* the better it gets at its actual job.

The defect: $\mathbb{P}(\rho_C > 1) = 1.0$ in both runs.

	w256	w768	classical best
μ_F (complement)	0.5803	0.1456	SymGS 0.1889
μ_C (coarse)	1.0400	1.0928	SymGS 0.4695
$\max \rho_C$	1.0837	1.1970	—
$\mathbb{P}(\rho_C > 1)$	1.00	1.00	0.00
μ_C/μ_F	1.79	7.50	2.49

The learned step is **the only method in the table that grows the coarse component**, and better complement reduction came with *worse* coarse behaviour. Every classical smoother reduces it, because a classical smoother is a contraction on the whole space. The network's objective never saw e_C , so its behaviour there is an uncontrolled extrapolation — and it turns out to be expansive rather than zero. This does not destroy selectivity ($\mu_C/\mu_F = 7.50$), but it is not nothing, and it is visible *only because the two components are measured separately*: a protocol reporting a single overall reduction factor would have hidden it completely.

What Is Established, and What Comes Next

The error split is exact and verified. $e = e_C + e_F$ with $e_C^T A e_F = 0$ and $P^T A e_F = 0$ hold to $\sim 10^{-15}$ on the actual hierarchy, asserted by the program rather than assumed.

They fix what each operation of the cycle is responsible for, and they make the first experiment an *extrapolation test*.

The loss is a modelling decision with a defensible answer. The damping objective is the one a single-level experiment can *interpret*; the energy norm is the one the iteration contracts in; the weighting makes the optimised metric and the reported metric the same; homogeneity and the fixed point put the network in the same class as its comparators.

The learned step does learn the smoothing property, and the two capacity diagnostics are what make that readable. With a branch of the complement's dimension (768): $\mu_F = 0.1456$, $\mu_C = 1.0928$, $\mu_C/\mu_F = 7.50$ — first of six, better than symmetric Gauss–Seidel, no amplified sample. With the trunk provably non-binding and the rank-768 linear limit at 0.0000, the residual is an *optimisation gap*, not a representational one.

And one defect is stated plainly: every coarse sample is amplified, by up to 19.7%, and the better the complement reduction the worse the amplification.

The protocol, fixed now, before the experiment: (i) the same hierarchy, test errors and norm; (ii) free parameters selected on validation, scored on test; (iii) the baseline re-run inside the same cycle; (iv) the *distribution* reported, not only the mean; and (v) the capacity bound published next to the result. Items (iv) and (v) exist because this work *needed* them: a run at the wrong branch width produced a plausible and wrong conclusion that only the distribution exposed.

The sequence of next experiments

#	Experiment	Why it is next
1	Stage II: the leak λ and the two-grid propagator	Needs <i>no</i> new training and no new data — the same test errors and weights answer it, because $E_{\text{TG}} = S_{\text{post}} \Pi_F S_{\text{pre}}$ exactly. It also <i>checks Stage I</i> , since the one-sided factor must reproduce $\mu_F = 0.1456$.
2	Re-tune per level; grow p with n_{dof}	μ_F at width 768 is a statement about <i>one</i> level; the complement dimension changes with refinement, so the width that can represent the exact correction changes too.
3	The coarse-space term in the loss	Now that the leak is a <i>measured</i> quantity, adding a penalty for it is a different experiment from adding one that was never measured.
4	Stage III: the complete V-cycle	Only after 1–3 have fixed the design: a cycle result cannot attribute a gain to a node the earlier stages have not isolated.

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